

Robust Local Collision Avoidance via Stochastic Vector Field Inequalities

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I. INTRODUCTION

THERE are many approaches to collision avoidance, which involve both global planning using rapidly exploring random trees (RRTs) [1], probabilistic roadmaps (PRMs), as well as control-based approaches such as artificial potential fields (APFs) [2] and control barrier functions (CBFs) [3]. RRTs generate random trees to explore the environment to connect the start and goal configurations and generates collision-free paths, with extensions such as particle RRTs for handling uncertainty in the environment [4]. PRMs generate a sampling-based roadmap that captures the connectivity of the free configuration space; however, they typically handle static workspaces and require preprocessing to generate the roadmap [1]. Despite APFs suffering from oscillations, local minima, and unreachable targets, they are still widely used in planning and control applications [2], mostly due to their simplicity. Conversely, CBFs achieve local collision avoidance by enforcing set invariance on the controller's input, this ensures that the system remains within the safe set [3]. They have been applied across the control field for collision avoidance, most notably in adaptive cruise control [5], bipedal robotic walking [6], and manipulation [7].

Vector field inequalities (VFIs) [8] can be considered a special case of CBFs [3, remark 4] where task-space constraints are derived via a geometric approach that, by construction, generate the forward-invariant safe set [8]. A whole-body controller for a holonomic mobile manipulator was implemented using VFIs in [9] to avoid obstacles and maintain visibility of a moving object. Second-order VFIs were introduced in [10] and applied to self-collision avoidance between elements of the robotic system. In [11], an adaptive controller was derived to account for kinematic inaccuracies in the model and adapt the system parameters online using VFIs used to enforce collision avoidance under control inputs and adaptation signals in the estimated task space.

The aforementioned approaches assume deterministic knowledge of the geometric primitives within the environment; as such, they only provide formal safety guarantees under these assumptions. Derwent et al. [12] addressed the discretisation error by buffering the constraints to ensure that system behaviour remained safe. In contrast, the CBF field addresses this problem by redesigning the controller in discrete time [13], [14] and extends it to stochastic systems formulated

in both discrete and continuous time. Reciprocal and zero CBFs for continuous-time stochastic systems was presented in [15]. Meanwhile, [16] investigated stochastic CBFs under state estimation for systems that evolve on manifolds and demonstrated the approach for systems evolving in $SE(2)$ and $SE(3)$. A probabilistic CBF framework for discrete-time stochastic systems was initially presented in [17], which reformulated the CBF as chance constraints. Chance constraints have also been widely used in stochastic model predictive control [18], with most research focusing on the effects of uncertainty and noise in the state variable.

A. Statement of Contributions

The contribution of this paper is the concept of stochastic VFIs (SVFIs), as being a stochastic extension of the VFI framework [8] that accounts for uncertainty in geometric primitives in the workspace, such as obstacles. This is achieved by introducing chance constraints into the constrained optimal task-space controller, thereby providing probabilistic guarantees of adhering to the desired distance dynamics and, by extension, collision avoidance with probabilistic guarantees. We implement SVFIs via a deterministic surrogate, which can be easily implemented as a second order cone constraint in the constrained optimal controller formulations. We provide a general formulation of the SVFIs and two specific examples of point-to-point and point-to-plane collision avoidance.

II. MATHEMATICAL BACKGROUND

A. Kinematic Control

The task-space kinematics of a mobile manipulator can be described as

$$\mathbf{x} = \mathbf{h}(\mathbf{q}), \quad (1)$$

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\mathbf{u}, \quad (2)$$

where $\mathbf{q} \triangleq \mathbf{q}(t) \in \mathbb{R}^n$ is the configuration vector of the robot, $\mathbf{x} \in \mathbb{R}^m$ is the task-space vector, $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function that maps the configuration vector to the task-space vector through the forward kinematics, $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{m \times n}$ is the task Jacobian matrix, and $\mathbf{u}$ is the control input given as configuration velocities (i.e., $\mathbf{u} \triangleq \dot{\mathbf{q}}$).

The desired deterministic optimal control law that enforces the task-space error to converge to zero exponentially as best as possible is formulated as a constrained quadratic program (QP) [8]. At any instance of time t , the measured $\mathbf{q}$ is used to solve the QP given by

$$\begin{aligned} \mathbf{u} \in \underset{\dot{\mathbf{q}}}{\operatorname{argmin}} \quad & \|\mathbf{J}(\mathbf{q})\dot{\mathbf{q}} + \kappa_t \tilde{\mathbf{x}}\|_2^2 + \lambda_c \|\dot{\mathbf{q}}\|_2^2 \\ \text{subject to} \quad & \mathbf{W}_i(\mathbf{q})\dot{\mathbf{q}} \preceq \mathbf{w}_i(\mathbf{q}), \end{aligned} \quad (3)$$

where $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$ is the task-space error, with $\dot{\mathbf{x}}_d = \mathbf{0}$ for all t ,

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and controller gain and damping are given by $\kappa_t, \lambda_c \in (0, \infty)$, respectively. The system is subject to a set of linear differential inequalities in the control inputs, in which $\mathbf{W}_i(\mathbf{q})$ and $\mathbf{w}_i(\mathbf{q})$ enforce task-space and configuration-space constraints through deterministic VFIs.

In practice, the system is subject to noise in the estimation of the configuration vector and geometric primitives that constrain the system. Given an estimate of the task- and configuration-space variables, the controller described in (3) can be rewritten as a stochastic optimal task-space controller given by

$$\begin{aligned} \mathbf{u} \in \operatorname{argmin}_{\mathbf{q}} \quad & \mathbb{E} \left[\|\mathbf{J}(\mathbf{q})\dot{\mathbf{q}} + \kappa_t \tilde{\mathbf{x}}\|_2^2 \right] + \lambda_c \|\dot{\mathbf{q}}\|_2^2 \\ \text{subject to} \quad & \mathbf{W}_i(\mathbf{q})\dot{\mathbf{q}} \preceq \mathbf{w}_i(\mathbf{q}) \\ & \Pr(\mathbf{W}_c(\mathbf{q})\dot{\mathbf{q}} \preceq \mathbf{w}_c(\mathbf{q})) \succeq \alpha_c, \end{aligned} \quad (4)$$

where additional SVFIs, formulated through chance constraints, are used to enforce a probability of at least α_c that the distance dynamics are satisfied.

B. Vector Field Inequalities

VFIs are a geometrical approach to CBF that enables the transformation of non-linear constraints in the robot configuration into linear differential inequalities in the control inputs. This approach is useful in enforcing collision avoidance by imposing a desired distance dynamics between two geometric primitives, where one or more are kinematically coupled to the robot. They leverage the signed distance and the squared distance between the primitives and the distance Jacobian that relates the time derivative of the distance to the robot's configuration velocities. The reader is directed to [8] for the general derivation of the constraints and implementation. For completeness, the essentials are summarised herein.

VFIs can enforce two behaviours, which are staying within a safe zone or outside of a restricted zone with a (constant) minimum safety distance $d_{\text{safe}} \in [0, \infty)$. Given $d \triangleq d(\mathbf{q})$ as the distance between two primitives, where one is kinematically coupled to the robot, then the distance to the boundary is given by $\tilde{d} \triangleq \tilde{d}(\mathbf{q}) = d - d_{\text{safe}}$. The restricted zone Ω_R and the safe zone Ω_S are given as

$$\Omega_R \triangleq \left\{ \mathbf{q} \in \mathbb{R}^n : \tilde{d}(\mathbf{q}) < 0 \right\}, \quad \Omega_S \triangleq \left\{ \mathbf{q} \in \mathbb{R}^n : \tilde{d}(\mathbf{q}) \geq 0 \right\}.$$

The signed distance dynamics can be expressed as

$$\dot{\tilde{d}} = \dot{d} - \dot{d}_{\text{safe}} = \dot{d}, \quad (5)$$

where a positive $\dot{\tilde{d}}$ means the robot is moving away from the boundary, whereas a negative $\dot{\tilde{d}}$ means the robot is approaching the boundary. These dynamics are constrained by

$$\dot{\tilde{d}} \geq -\eta \tilde{d} \iff -\mathbf{J}_d \dot{\mathbf{q}} \leq \eta \tilde{d}, \quad (6)$$

where $\mathbf{J}_d \triangleq \mathbf{J}_d(\mathbf{q}) \in \mathbb{R}^{1 \times n}$ is the distance Jacobian relating the primitive attached to the robot to the workspace primitive, and $\eta \in [0, \infty)$ is the maximum allowable approach rate to the boundary.

III. STOCHASTIC VECTOR FIELD INEQUALITIES

Stochastic VFIs (SVFIs) are a probabilistic extension to VFIs, similar to Stochastic CBFs, which takes into account the uncertainty in geometric primitives and provide probabilistic

guarantees of adhering to the desired distance dynamics. Let us consider a chance constraint that imposes a probability that a constraint is satisfied and is given by the general form

$$\Pr(g(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho}) \geq 0) \geq \alpha, \quad (7)$$

where $g(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho}) : \mathbb{R}^w \times \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is the function describing the distance dynamics, in which $\boldsymbol{\chi} \in \mathbb{R}^w$ is the vector of deterministic variables, $\mathbf{u} \in \mathbb{R}^n$ is the control input vector, and $\boldsymbol{\rho} \in \mathbb{R}^m$ is the stochastic parameters. The parameter $\alpha \in (0.5, 1)$ is the confidence interval that gives the probability that the constraint is satisfied. We use (6) in (7) to obtain a general stochastic form of a SVFI as

$$\Pr(\dot{d}(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho}) + \eta d(\boldsymbol{\chi}, \boldsymbol{\rho}) \geq 0) \geq \alpha,$$

where $\dot{d}(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho})$ and $d(\boldsymbol{\chi}, \boldsymbol{\rho})$ follow different probability distributions related to the distance between two primitives.

Let us assume that the distribution of the stochastic parameters is Gaussian and described as $\boldsymbol{\rho} \sim \mathcal{N}(\bar{\boldsymbol{\rho}}, \boldsymbol{\Sigma}_\rho)$. Since the distribution is Gaussian, there exists a deterministic surrogate that can be used within the deterministic optimisation problem. Although, the function $g(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho})$ may not be linear in general, the final distribution can be approximated by using a first order Taylor series of the function and propagating the covariance through the linearisation. Applying this to the distance dynamics, we get $g(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho}) = \dot{d}(\boldsymbol{\chi}, \mathbf{u}, \boldsymbol{\rho}) + \eta d(\boldsymbol{\chi}, \boldsymbol{\rho})$, where we assume $g \sim \mathcal{N}(\bar{g}, \sigma_g^2)$. The mean distance dynamics, $\bar{g} = g|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}}$, is the expected value of g evaluated using the expected stochastic variables, and the variance of the distance dynamics is given by the linear transformation of the covariance [19],

$$\sigma_g^2 = \left. \frac{\partial g}{\partial \boldsymbol{\rho}} \right|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}} \boldsymbol{\Sigma}_\rho \left. \frac{\partial g}{\partial \boldsymbol{\rho}} \right|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}}^T, \quad (8)$$

where $\frac{\partial g}{\partial \boldsymbol{\rho}}$ is the partial derivative of g with respect to each stochastic variable in $\boldsymbol{\rho}$. Given that the covariance matrix is strictly symmetric positive-definite, the Cholesky decomposition of the covariance matrix is given as $\boldsymbol{\Sigma}_\rho = \mathbf{L}_\rho \mathbf{L}_\rho^T$, where $\mathbf{L}_\rho$ is lower triangular. Then, we can rewrite (8) in terms of the squared Euclidean norm as

$$\sigma_g^2 = \left. \frac{\partial g}{\partial \boldsymbol{\rho}} \right|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}} \mathbf{L}_\rho \mathbf{L}_\rho^T \left. \frac{\partial g}{\partial \boldsymbol{\rho}} \right|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}}^T = \left\| \mathbf{L}_\rho^T \left. \frac{\partial g}{\partial \boldsymbol{\rho}} \right|_{\boldsymbol{\rho}=\bar{\boldsymbol{\rho}}} \right\|_2^2. \quad (9)$$

To convert this to a deterministic surrogate, we first scale it to a standard normal random variable, $Z \sim \mathcal{N}(0, 1)$, by

$$\Pr\left(\underbrace{\frac{g - \bar{g}}{\sigma_g}}_Z \geq \frac{0 - \bar{g}}{\sigma_g}\right) \geq \alpha \iff 1 - \Phi\left(\frac{0 - \bar{g}}{\sigma_g}\right) \geq \alpha, \quad (10)$$

where Φ is the standard Gaussian cumulative density function [20]. This allows us to use the inverse cumulative distribution function, Φ^{-1} , on (10).

Noting that $\Phi(-x) = 1 - \Phi(x)$ and that Φ is a monotonically increasing continuous function, we obtain

$$\Phi\left(\frac{\bar{g}}{\sigma_g}\right) \geq \alpha \iff \bar{g} \geq \sigma_g \Phi^{-1}(\alpha). \quad (11)$$

By substituting (9) into (11), the final deterministic surrog-

ate for the SVFI is given as

$$\bar{g} \geq \Phi^{-1}(\alpha) \left\| \mathbf{L}_\rho^T \frac{\partial g}{\partial \rho} \right\|_{\rho=\bar{\rho}}^T, \quad (12)$$

which can be included into the original stochastic controller (4) in the form of a second order cone constraint as follows:

$$\begin{aligned} \mathbf{u} \in \underset{\dot{\mathbf{q}}}{\text{argmin}} \quad & \| \mathbf{J}(\bar{\mathbf{q}}) \dot{\mathbf{q}} + \kappa_t \check{\mathbf{x}} \|_2^2 + \lambda_c \| \dot{\mathbf{q}} \|_2^2 \\ \text{subject to} \quad & \mathbf{W}_i(\bar{\mathbf{q}}) \dot{\mathbf{q}} \preceq \mathbf{w}_i(\bar{\mathbf{q}}) \\ & \Phi^{-1}(\alpha) \left\| \mathbf{L}_\rho^T \frac{\partial g}{\partial \rho} \right\|_{\rho=\bar{\rho}}^T \preceq \bar{\mathbf{J}}_d \dot{\mathbf{q}} + \eta \bar{d}, \end{aligned} \quad (13)$$

where $\bar{\mathbf{q}}$ is the expected robot configuration, $\check{\mathbf{x}} = \bar{\mathbf{x}} - \mathbf{x}_d$ is the expected task-space error, $\bar{\mathbf{J}}_d \triangleq \bar{\mathbf{J}}_d(\bar{\mathbf{q}}) \in \mathbb{R}^{1 \times n}$ is the expected distance Jacobian relating the primitive attached to the robot to the expected workspace primitive, and $\bar{d}$ is the expected distance between the primitives evaluated at $\bar{\rho}$. The second constraint in Problem (13) is a second-order conic constraint because, as shown in Sections III-A and III-B, the term inside the norm can always be written as $\mathbf{A}\dot{\mathbf{q}} + \mathbf{b}$. Therefore, this optimization problem can be solved using conventional quadratic second-order conic program solvers.

A. Point-to-point SVFI

Let us consider a static point feature $\mathbb{R}^3 \ni \mathbf{p} \sim \mathcal{N}(\bar{\mathbf{p}}, \Sigma_p)$ in the workspace, then the squared distance between $\mathbf{p}$ and a point $\mathbf{t}(\mathbf{q}) \in \mathbb{R}^3$ on the robot's kinematic chain [8] is given by

$$D_{t,p} = \|\mathbf{t}(\mathbf{q}) - \mathbf{p}\|_2^2. \quad (14)$$

The desired distance dynamics can be defined as

$$\dot{D}_{t,p} \geq -\eta \bar{D}_{t,p} \iff 2(\mathbf{t}(\mathbf{q}) - \mathbf{p})^T \mathbf{J}_t \dot{\mathbf{q}} \geq -\eta \bar{D}_{t,p},$$

with $\bar{D}_{t,p} = D_{t,p} - d_{\text{safe}}^2$ and $\mathbf{J}_t = \partial \mathbf{t} / \partial \mathbf{q}$. Consider $g_{t,p} \triangleq D_{t,p} + \eta \bar{D}_{t,p}$, then we assume $g_{t,p} \sim \mathcal{N}(\bar{g}_{t,p}, \sigma_{t,p}^2)$, where $\bar{g}_{t,p} = g_{t,p}|_{\mathbf{p}=\bar{\mathbf{p}}}$ and $\sigma_{t,p}$ is the variance in the distance dynamics. Thus, we use (8) to obtain

$$\begin{aligned} \sigma_{t,p}^2 &= \left. \frac{\partial g_{t,p}}{\partial \mathbf{p}} \right|_{\mathbf{p}=\bar{\mathbf{p}}} \Sigma_p \left. \frac{\partial g_{t,p}}{\partial \mathbf{p}} \right|_{\mathbf{p}=\bar{\mathbf{p}}}^T, \\ \left. \frac{\partial g_{t,p}}{\partial \mathbf{p}} \right|_{\mathbf{p}=\bar{\mathbf{p}}} &= -2\dot{\mathbf{q}}^T \mathbf{J}_t^T - 2\eta(\mathbf{t}(\bar{\mathbf{q}}) - \bar{\mathbf{p}})^T. \end{aligned} \quad (15)$$

By using the Cholesky decomposition, $\Sigma_p = \mathbf{L}_p \mathbf{L}_p^T$, where $\mathbf{L}_p \in \mathbb{R}^{3 \times 3}$, and using (15), we obtain $\sigma_{t,p}^2 = \left\| 2\mathbf{L}_p^T (\mathbf{J}_t \dot{\mathbf{q}} + \eta(\mathbf{t}(\bar{\mathbf{q}}) - \bar{\mathbf{p}})) \right\|_2^2$. Using (12), (14), and (15), with $\bar{D}_{t,p} = D_{t,p}|_{\mathbf{p}=\bar{\mathbf{p}}}$, the deterministic surrogate constraint is given by

$$\begin{aligned} \Phi^{-1}(\alpha) \left\| 2\mathbf{L}_p^T \mathbf{J}_t \dot{\mathbf{q}} + 2\eta \mathbf{L}_p^T (\mathbf{t}(\bar{\mathbf{q}}) - \bar{\mathbf{p}}) \right\|_2 &\leq \\ 2(\mathbf{t}(\bar{\mathbf{q}}) - \bar{\mathbf{p}})^T \mathbf{J}_t \dot{\mathbf{q}} + \eta (\bar{D}_{t,p} - d_{\text{safe}}^2). \end{aligned} \quad (16)$$

B. Point-to-plane SVFI

Following the closest point representation in [21], a static plane not passing through the origin can be given by $\boldsymbol{\pi} = d_\pi \mathbf{n}_\pi$, where $d_\pi = \|\boldsymbol{\pi}\|_2 \in \mathbb{R}$ is the signed perpendicular distance between the plane and the origin of the reference frame and $\mathbf{n}_\pi = \boldsymbol{\pi} / d_\pi \in \mathbb{S}^2$ is the normal of the plane. The

distribution of the plane is given by $\boldsymbol{\pi} \sim \mathcal{N}(\bar{\boldsymbol{\pi}}, \Sigma_\pi)$, with $\bar{\boldsymbol{\pi}} = \bar{d}_\pi \bar{\mathbf{n}}_\pi$, and $\Sigma_\pi \in \mathbb{R}^{3 \times 3}$ being the covariance matrix associated with the plane estimation. Given an arbitrary point $\mathbf{t}(\mathbf{q}) \in \mathbb{R}^3$ attached to the robot, the point-to-plane distance with respect to $\boldsymbol{\pi}$ is given by

$$d_{t,\pi}^\pi = \langle \mathbf{n}_\pi, \mathbf{t}(\mathbf{q}) \rangle - d_\pi. \quad (17)$$

The desired distance dynamics, enforced through the use of VFIs [8], is expressed as

$$\dot{d}_{t,\pi}^\pi \geq -\eta \bar{d}_{t,\pi}^\pi \iff \mathbf{n}_\pi^T \mathbf{J}_t \dot{\mathbf{q}} \geq -\eta \bar{d}_{t,\pi}^\pi, \quad (18)$$

where $\bar{d}_{t,\pi}^\pi = (d_{t,\pi}^\pi - d_{\text{safe}})$. Let $g_{t,\pi} \triangleq \dot{d}_{t,\pi}^\pi + \eta \bar{d}_{t,\pi}^\pi$, where we assume $g_{t,\pi} \sim \mathcal{N}(\bar{g}_{t,\pi}, \sigma_{t,\pi}^2)$, with $\bar{g}_{t,\pi} = g_{t,\pi}|_{\boldsymbol{\pi}=\bar{\boldsymbol{\pi}}}$ and $\sigma_{t,\pi}$ being the variance in the distance dynamics. Therefore,

$$\sigma_{t,\pi}^2 = \left. \frac{\partial g_{t,\pi}}{\partial \boldsymbol{\pi}} \right|_{\boldsymbol{\pi}=\bar{\boldsymbol{\pi}}} \Sigma_\pi \left. \frac{\partial g_{t,\pi}}{\partial \boldsymbol{\pi}} \right|_{\boldsymbol{\pi}=\bar{\boldsymbol{\pi}}}^T, \quad \text{with} \quad (19)$$

$$\begin{aligned} \left. \frac{\partial g_{t,\pi}}{\partial \boldsymbol{\pi}} \right|_{\boldsymbol{\pi}=\bar{\boldsymbol{\pi}}} &= \left(\frac{\partial g_{t,\pi}}{\partial \mathbf{n}_\pi} \frac{\partial \mathbf{n}_\pi}{\partial \boldsymbol{\pi}} + \frac{\partial g_{t,\pi}}{\partial d_\pi} \frac{\partial d_\pi}{\partial \boldsymbol{\pi}} \right) \Big|_{\substack{\mathbf{n}_\pi=\bar{\mathbf{n}}_\pi \\ d_\pi=\bar{d}_\pi}} \\ &= \left[\dot{\mathbf{q}}^T \mathbf{J}_t^T + \eta \mathbf{t}(\bar{\mathbf{q}})^T \quad -\eta \right] \underbrace{\begin{bmatrix} \bar{d}_\pi^{-1} (\mathbf{I} - \bar{\mathbf{n}}_\pi \bar{\mathbf{n}}_\pi^T) \\ \bar{\mathbf{n}}_\pi^T \end{bmatrix}}_{\mathbf{M} \in \mathbb{R}^{4 \times 3}} \end{aligned} \quad (20)$$

(21)

where the last equality holds because

$$\begin{aligned} \frac{\partial g_{t,\pi}}{\partial \mathbf{n}_\pi} &= \dot{\mathbf{q}}^T \mathbf{J}_t^T + \eta \mathbf{t}(\bar{\mathbf{q}})^T, & \frac{\partial \mathbf{n}_\pi}{\partial \boldsymbol{\pi}} &= \frac{\mathbf{I}_{3 \times 3} - \mathbf{n}_\pi \mathbf{n}_\pi^T}{d_\pi}, \\ \frac{\partial g_{t,\pi}}{\partial d_\pi} &= -\eta, & \frac{\partial d_\pi}{\partial \boldsymbol{\pi}} &= \mathbf{n}_\pi^T. \end{aligned}$$

Using Cholesky decomposition, we have $\Sigma_\pi = \mathbf{L}_\pi \mathbf{L}_\pi^T$. Then, using (9) and (21), we obtain

$$\sigma_{t,\pi}^2 = \left\| \mathbf{L}_\pi^T \mathbf{M}^T \begin{bmatrix} \mathbf{J}_t \dot{\mathbf{q}} + \eta \mathbf{t}(\bar{\mathbf{q}}) \\ -\eta \end{bmatrix} \right\|_2^2,$$

where $\boldsymbol{\Omega}^T \triangleq \mathbf{L}_\pi^T \mathbf{M}^T = \begin{bmatrix} \mathbf{J}_t^T & \boldsymbol{\Omega}_\eta^T \end{bmatrix} \in \mathbb{R}^{3 \times 4}$, with $\boldsymbol{\Omega}_t^T \in \mathbb{R}^{3 \times 3}$ and $\boldsymbol{\Omega}_\eta^T \in \mathbb{R}^{3 \times 1}$. This allows the variance to be written as

$$\sigma_{t,\pi}^2 = \left\| \boldsymbol{\Omega}_t^T \mathbf{J}_t \dot{\mathbf{q}} + \eta \boldsymbol{\Omega}_t^T \mathbf{t}(\bar{\mathbf{q}}) - \eta \boldsymbol{\Omega}_\eta^T \right\|_2^2. \quad (22)$$

and utilising (12), (17), and (22), the final deterministic surrogate constraint is given by

$$\begin{aligned} \Phi^{-1}(\alpha) \left\| \boldsymbol{\Omega}_t^T \mathbf{J}_t \dot{\mathbf{q}} + \eta \boldsymbol{\Omega}_t^T \mathbf{t}(\bar{\mathbf{q}}) - \eta \boldsymbol{\Omega}_\eta^T \right\|_2 &\leq \\ \bar{\mathbf{n}}_\pi^T \mathbf{J}_t \dot{\mathbf{q}} + \eta (\bar{d}_{t,\pi}^\pi - d_{\text{safe}}). \end{aligned} \quad (23)$$

IV. SIMULATIONS AND DISCUSSIONS

In all simulations, the control law described in (13) was implemented using DQ Robotics [22]. The measured robot configuration was noiseless; therefore, only the estimated primitives are noisy. The chosen solver used was MOSEK [23] with a custom interface created to integrate with the DQ robotics library. The CoppeliaSim [24] simulation was executed on an Intel Core i7 laptop with eight cores and a maximum frequency of 4.90 GHz, with 16 GB of RAM.

Table I: Comparison of the comp. time with a single constraint.

	Inequality computation time (μs)			MOSEK solve time of control law (13) (μs)		
	Min	Avg	Max	Min	Avg	Max
Ineq. (16)	104.65	132.65	379.1	701.72	1181.5	7505.0
Ineq. (23)	63.2	71.34	175.3	701.2	868.0	2259.2

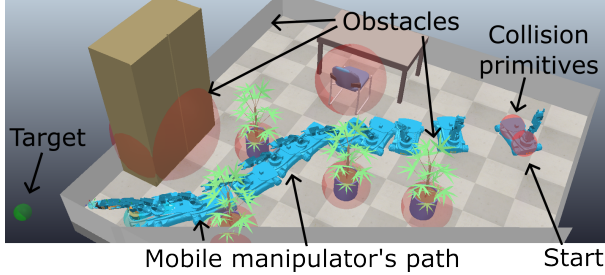


Figure 1: The mobile manipulator's path through the workspace whilst the system is performing local collisions with the estimated spheres enclosing six obstacles in the environment and the four walls bounding the workspace.

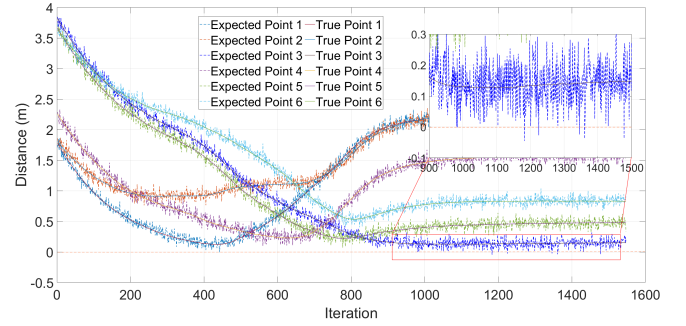
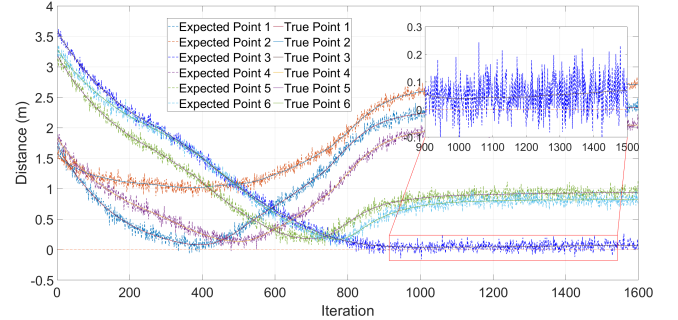
A. Computation cost

The evaluation of the computational time of the proposed control law was performed over a total of 12 simulations for 1000 time-steps each using a free-flying point to represent the robot. Then the minimum, maximum and average were calculated for each simulation and then averaged across all simulations to give a representative minimum, maximum and average execution time for both generating the required terms of the SVFIs and the time to compute the quadratic second-order conic program (13), summarised in table I. In each case, the first solver iteration was removed due to the addition of the overhead in configuring and initialising the second-order conic model used by [23], which is not representative of the time requirements for each constraint.

B. Simulation of the mobile manipulator

To simulate a real world scenario, a scene was set up within CoppeliaSim 4.10 using the KUKA youBot mobile manipulator with several objects placed within the environment, such as chairs, plants and cupboards. Each was encased in a sphere with the position and covariance being estimated by 10 noisy samples that followed $p_i \sim \mathcal{N}(\bar{p}_i, \sigma_i \mathbf{I}_{3 \times 3})$, with $\sigma = 0.2$ and $i = 1, \dots, 10$. Five planes used to bound the workspace were estimated from a noisy point-cloud, with points following a normal distribution with $\sigma = 0.04 \mathbf{I}_{3 \times 3}$. Figure 1 shows the setup with the mobile manipulator's path through the environment with a safety distance between the robot and each primitive being the sum of their individual radii.

A total of 34 SVFIs were used during the simulation, with spheres attached at the robot's end effector with two spheres covering the mobile base. A value of $\alpha = 0.75$ was used throughout the simulation, with $\eta = 1$, $\kappa_t = 5$, and $\lambda_c = 0.8$ with a sampling time of 10 ms. The average solver time for each iteration in this scenario was 5.68 ms, with the distance between the mobile manipulator and each obstacle presented in figure 2a. The plots in figure 2a show that the robot

(a) $\alpha = 0.75$ (b) $\alpha = 0.5$, equivalent to deterministic VFIs because $\Phi^{-1}(0.5) = 0$ Figure 2: The expected and true point-to-point distances for one collision primitive on the robot's base to each of the six obstacles within the environment, where the probabilities of satisfying the SVFIs are given by α .

maintained a minimum safety distance to obstacles without any violations despite the noisy estimation of the primitives within the environment. Between 900 and 1500 iterations, the estimated distance violated the minimum safety distance numerous times, but the SVFIs prevented the boundary from ever being violated (the solid line never goes below zero). When deterministic VFIs are used, as in figure 2b, substantial more violations of the expected distance occur over the same period. Since the true distance can be measured, the actual minimum safety distance is not violated in either case. However, in a real world scenario, SVFIs provide a probabilistic safety guarantee, which can be adjusted according to the risk associated to the robot operation. In contrast, deterministic VFIs, which are equivalent to SVFIs when $\alpha = 0.5$, can only guarantee safety with a modest probability of 50%.

V. CONCLUSION AND FUTURE WORK

We have presented a method for generating robust constraints by introducing SVFIs to design a controller with probabilistic guarantees of satisfying the distance-dynamics constraints. This inclusion allows us to account for uncertainty in simple geometric primitives, thereby achieving robust local collision avoidance. Future work will look to expand this to other geometric primitives and to account for uncertainty in the configuration vector, which is particularly important for mobile bases. This approach can be extended to other methods, such as second-order VFIs [10], and combined with a state estimator and SLAM algorithms to generate covariance matrices online, allowing the system to traverse the environment with greater certainty.

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